Breaking the Black Box - Evals

Craig West

https://craig-west.netlify.app/

https://evaluating-ai-agents.com/

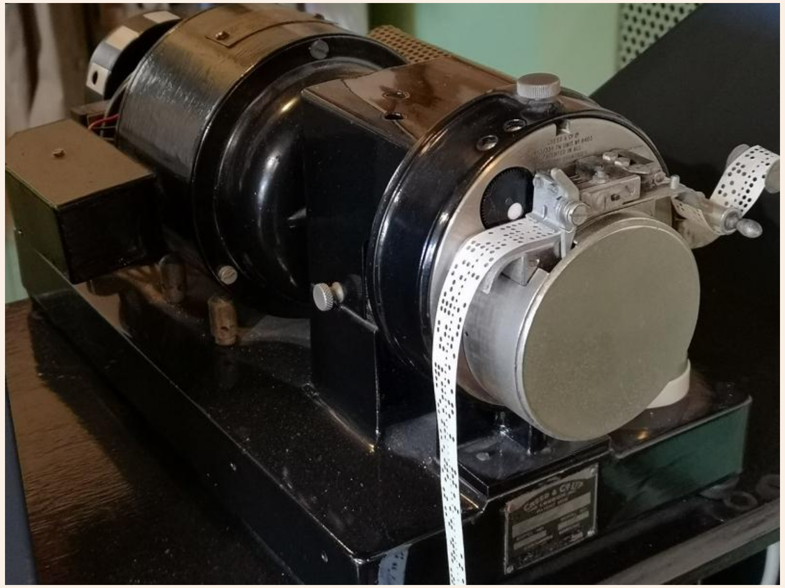
- Contains links for slides and eval-framework
- Everything in this talk is included

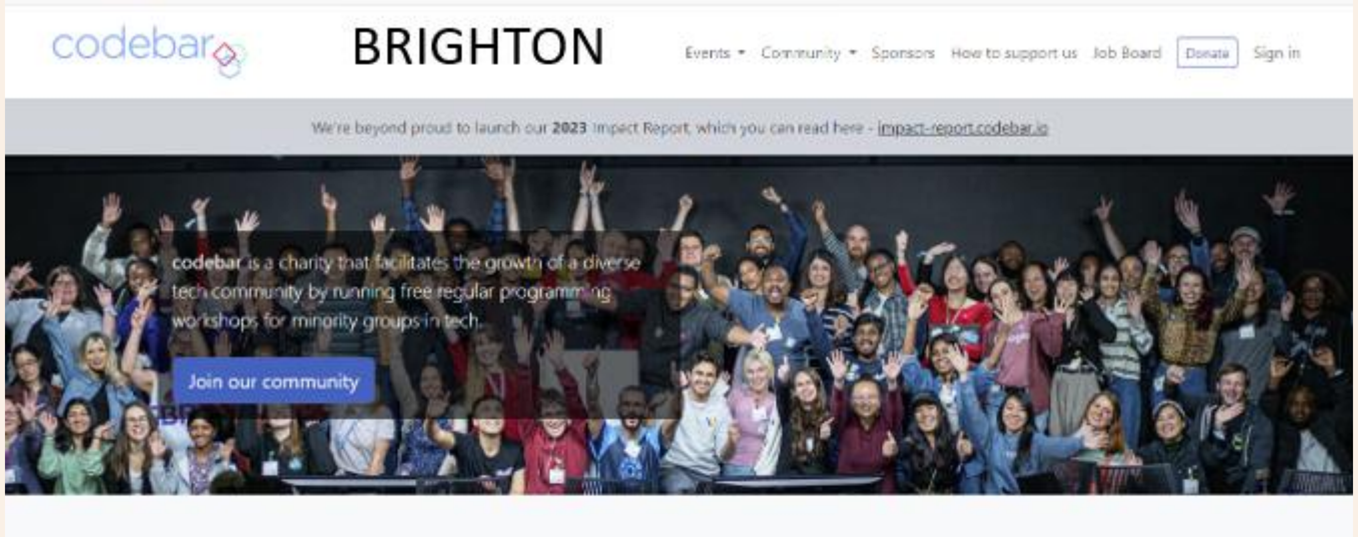
About me


Brighton
I live ~ one mile


Leo - Fox Red
Lab


Local Fox follows
us home

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We're beyond proud to launch our 2023 Impact Report, which you can read here - impact-report.codebar.io
codebar is a charity that facilitates the growth of a diverse tech community by running free regular programming workshops for minority groups in tech.
Join our community

My desired outcome

Not for you to fully understand Evals

But…

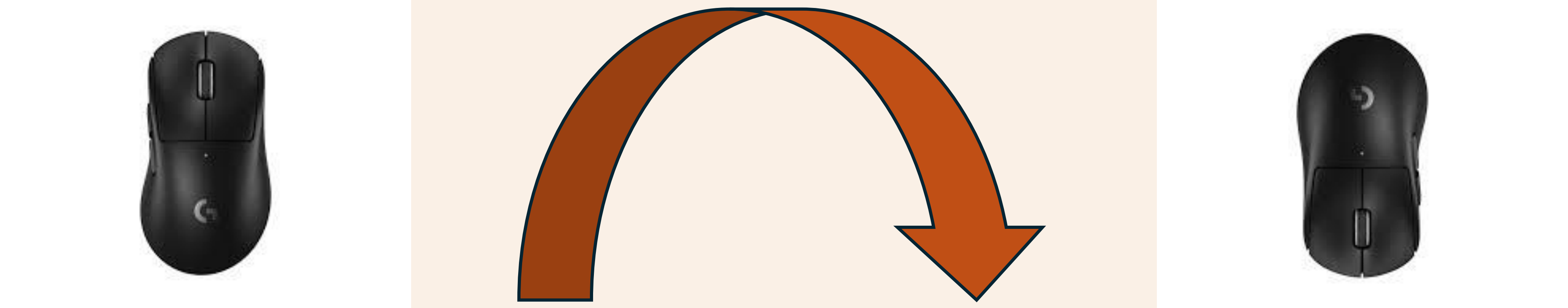
That Evals are doable

There are useful resources available for you

Demystify and simplify what an Agent is

There will be some slides and then we will dive into the code and manual (website).

180-degree shift



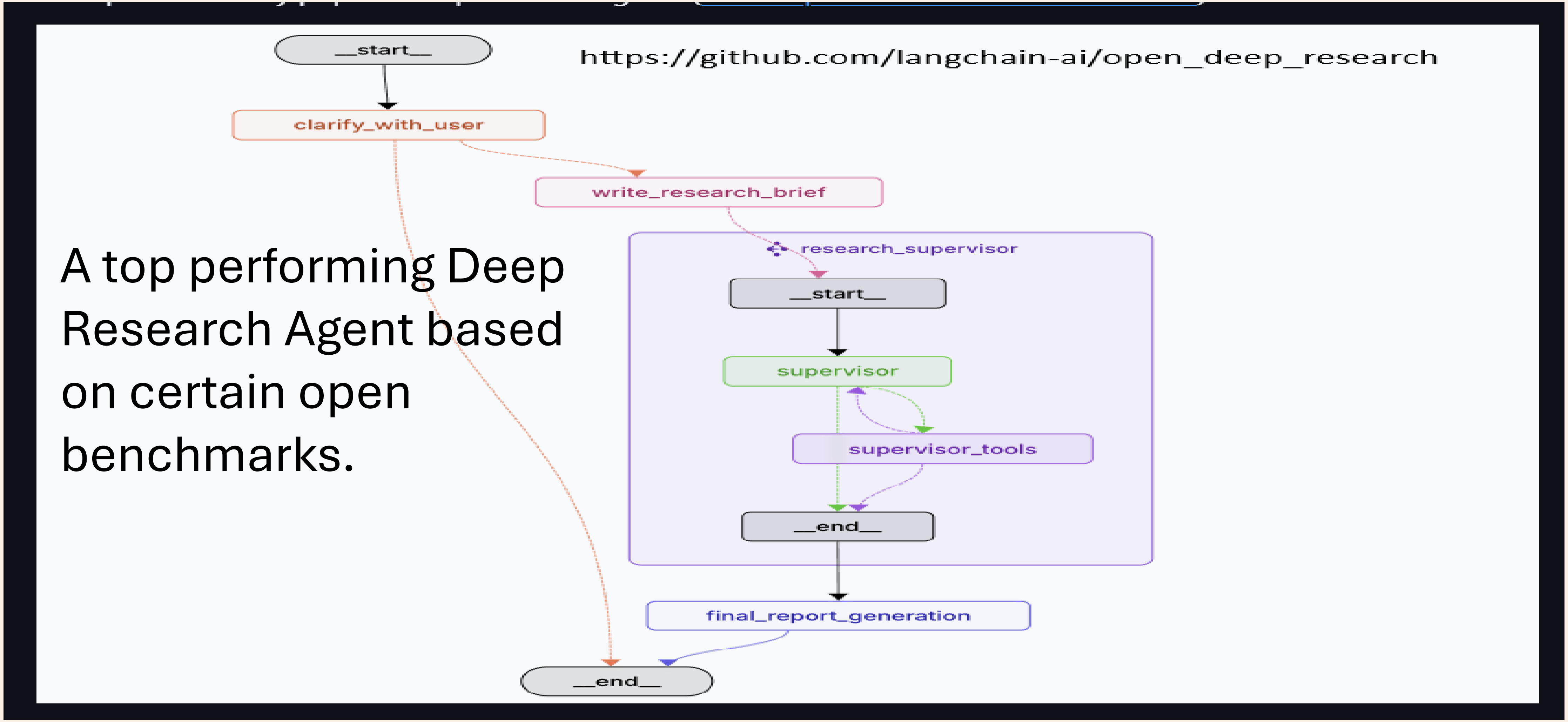
No more difficult than what we do currently – just 180 degrees different

“It doesn’t get any easier – just different” - Anon

But I don't use Agents!

- Perhaps not now, but it might be that your app is required to use Natural Language, (text, voice) from the user as the front end.
- You might use or incorporate ready made 3rd party software that is Agentic as in the next two slides…
- or use Model Context Protocol/Agent2Agent in your 'app'.
- Or need to make an MCP Server
- It may be helpful to understand what Agents are as they seem to crop up everywhere.

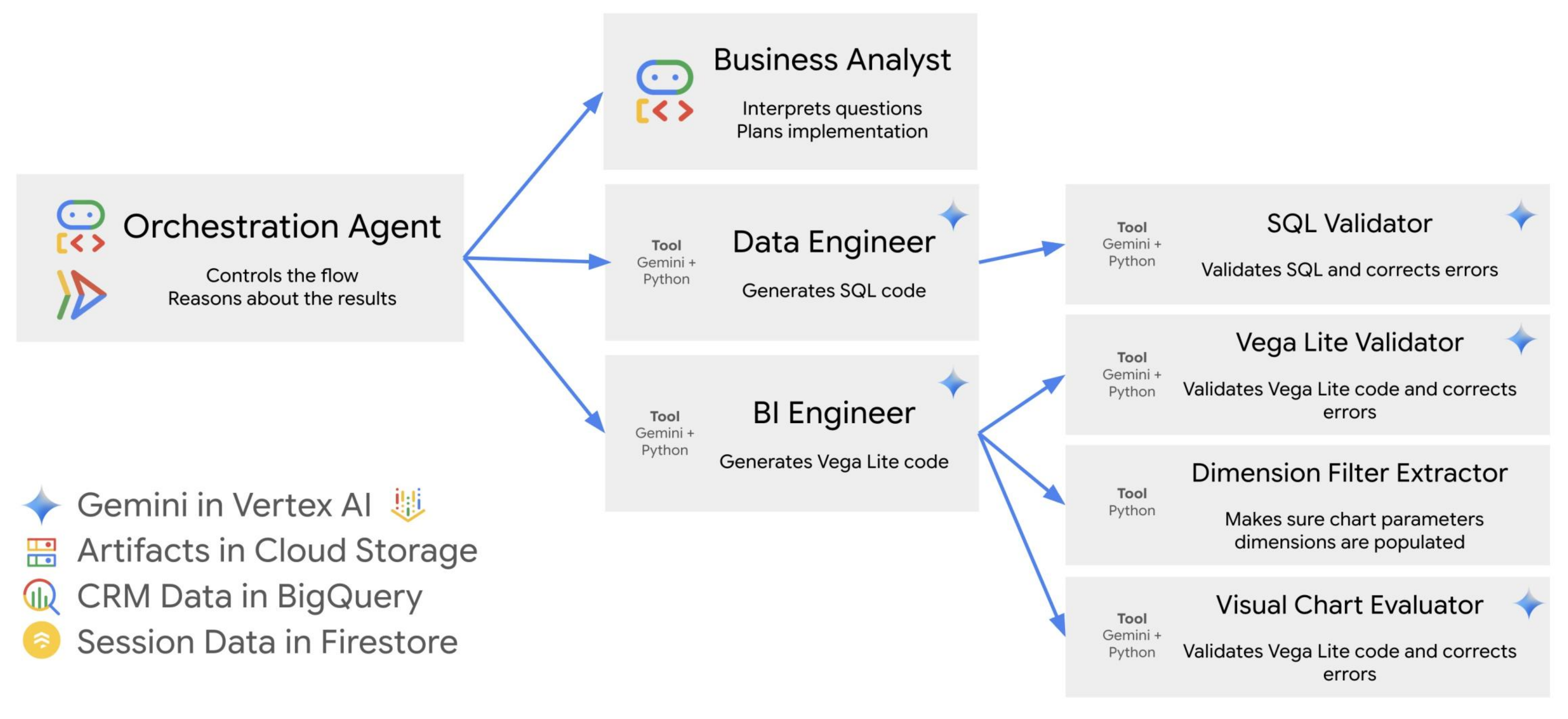
Langchain Deep Research Agent



A top performing Deep Research Agent based on certain open benchmarks.

https://github.com/langchain-ai/open_deep_research

Google CRM Agent



https://github.com/vladkol/crm-data-agent/blob/main/src/agents/data_agent/prompts/crm_business_analyst.py
https://www.youtube.com/watch?v=2-AJszogj7Y

Why Evals?

Ensure TRUST and CAPABILITIES of our AGENTS
so that somebody will pay for them.

Would you pay for a substandard product?

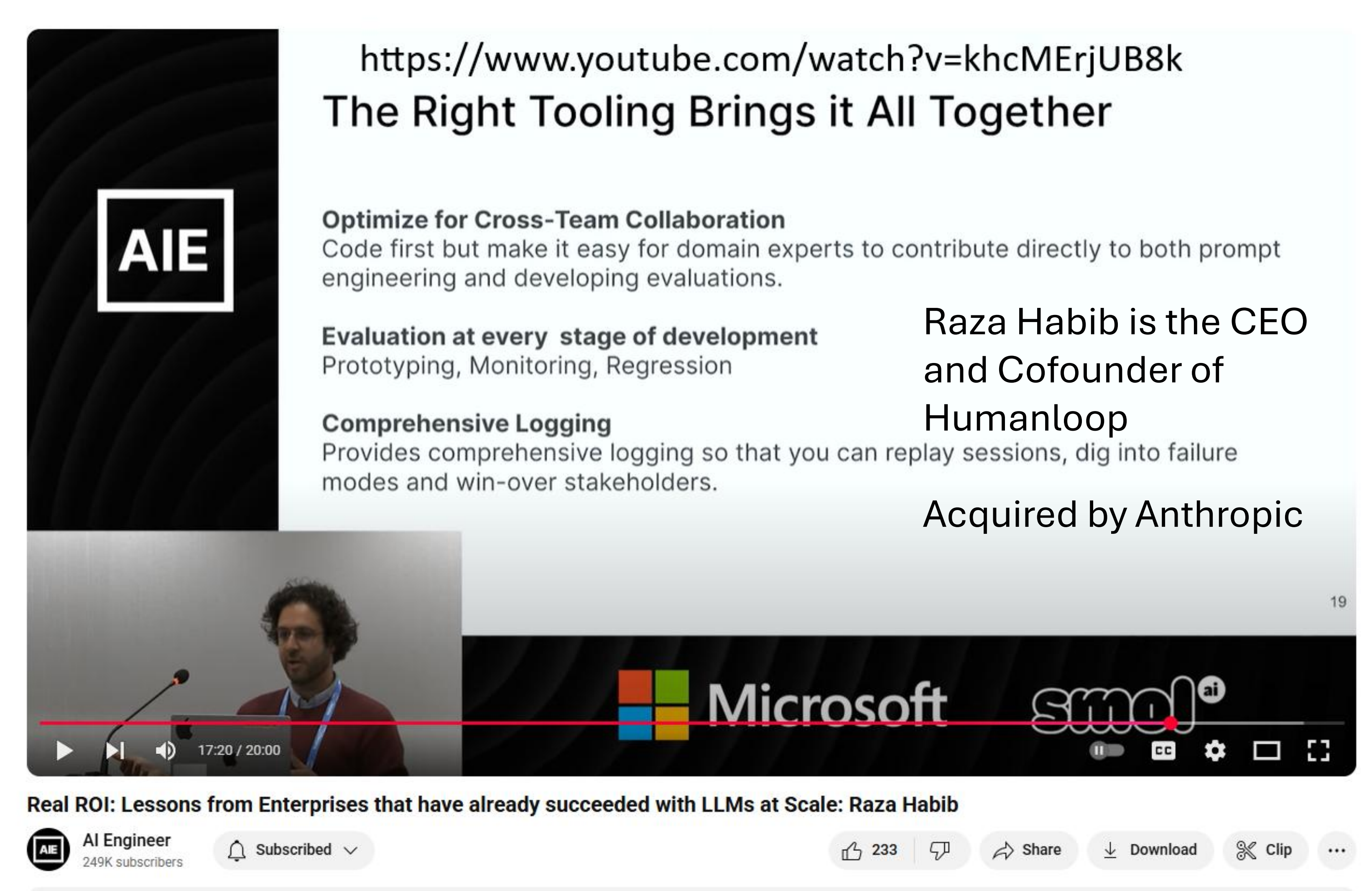
Excellence in an evaluator will rub off on developer prompt code.

Can we save £ if we use a different LLM?

Model costs

Cost of a unit of ‘intelligence has dropped ~ 99% in last two years and is 10x a year (anecdotal)

When to use Evals?



If we are to design and build an F1 race car, we might need a set of specs to develop, test and monitor against.

What is an AI Agent?

Many definitions and people opt for ‘Agentic Apps’

Anthropic

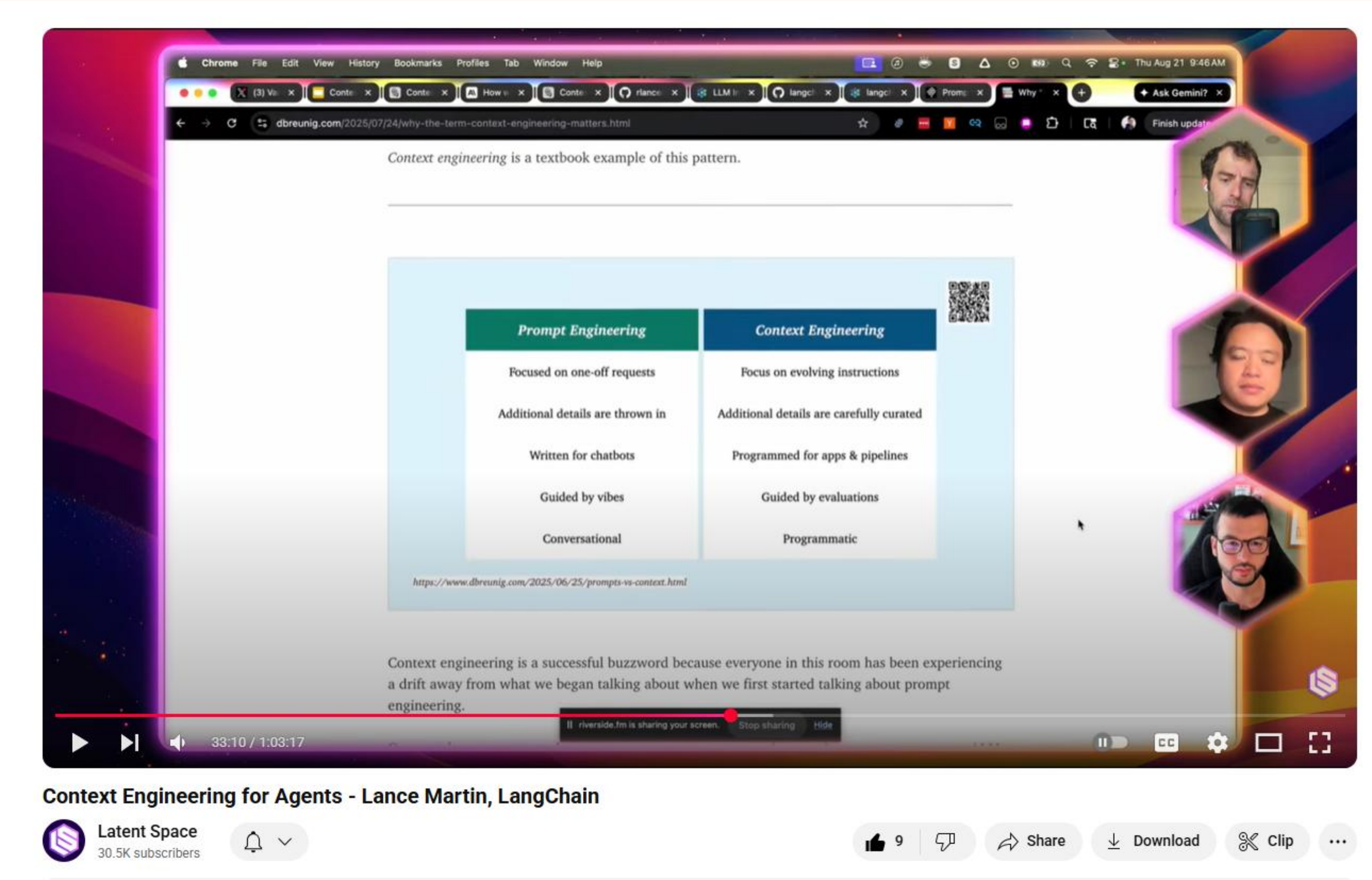
- **Workflows** are systems where LLMs and tools are orchestrated through predefined code paths.
- **Agents**, on the other hand, are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.

API

```
model = "gpt-4o-mini"  # Model
model_endpoint = "https://api.openai.com/v1/chat/completions"  # just one endpoint
headers = {
    "Content-Type": "application/json",  # Authorisation
    "Authorization": f"Bearer {api_key}",
}
# payload structure may vary from LLM Organisation but it is a text string.
payload = {
    "model": model,
    "messages": [
        {"role": "system", "content": system_prompt},
        {"role": "user", "content": user_prompt},
    ],
    # additional parameters
    "stream": False,
    "temperature": temperature,
}
# Use HTTP POST method
response = requests.post(
    url=model_endpoint,  # The endpoint we are sending the request to. Low Temperature:
    headers=headers,  # Headers for authentication etc
    data=json.dumps(payload),  # Inputs, Context, Instructions, Additonal parameters
).json()
```

These messages contain Instructions and Context

What is Prompt/Context Engineering?



What is Prompt/Context Engineering?

Prompt Engineering	*Context Engineering*
Focused on one-off requests	Focus on evolving instructions
Additional details are thrown in	Additional details are carefully curated
Written for chatbots	Programmed for apps & pipelines
Guided by vibes	Guided by evaluations
Conversational	Programmatic

https://www.dbreunig.com/2025/06/25/prompts-vs-context.html

Where to start?   What to do?

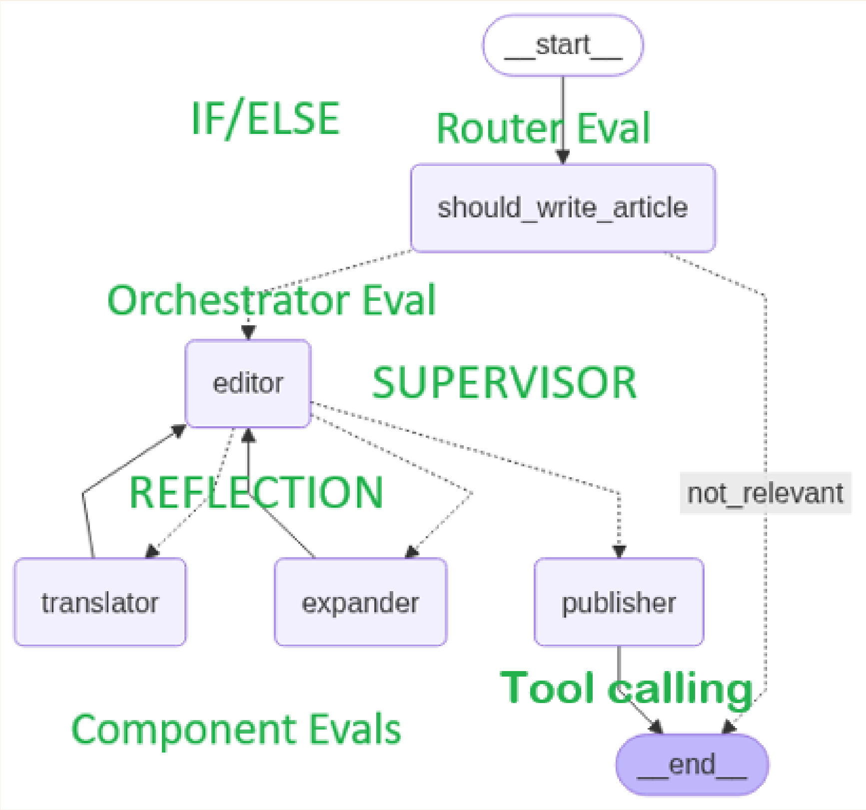
- We will look at Case Study 1 in the manual and framework.

Given an article title, determine if it is about AI.

1. Take title and create article of N words (EXPANDER)
2. In a specified language (TRANSLATOR)
3. Ensure article is not sensational (EDITOR)
4. Publisher determines prices with dependent tools (TOOL CALLING) – calculate price and then categorise this price.

GOTO PUBLISH = YES only if 1,2 and 3 are YES

This app could be a leaf of a larger app.



A possible workflow

We build our app with a basic prompt “Write an article on {TOPIC} of around {N} words in language {LANGUAGE}.”

We test manually during dev.

It all works mechanistically.

Now lets progress…

A possible workflow

We decide to add a simple 💛 to see what cusomers like.

We notice that users like:

Bullet points – Abstract - Conclusion

A possible workflow

We change our prompt to include these features.

However, only 2% of users give feedback so we decide to use Human Evaluators to give feedback as well as label articles as ‘good article’ etc.

A possible workflow

We need to draft up guidelines on what an article should contain, its structure and other ways an article is ‘good’.

This exercise further helps our LLM app prompt, and we refine it.

A possible workflow

We can only handle/afford a small percentage, so we decide to use an LLM as Judge.

Seems illogical but is actually very effective. An uncalibrated bitof code calibrating another

With our original prompt and our human evaluator guidelines we write a prompt for this LLM Judge. We can now process a much greater percentage online and offline.

It is better than our app! We refine our app prompt...

A possible workflow

Now that we are looking at our data, we see that sometimes the app offers financial advice.

We must prevent this (guardrails) so we create an LLM as Judge that detects if this article had financial advice.

It saves many problems...

A possible workflow

Now that we have a robust way of evaluating our app, we can start to tinker:

- Different models in different places for cost reduction yet same performance?
- Does one part of the app give greater benefit for unit of change than another?
- Can we reduce latency without breaking anything (regression testing)
- Offer a client dashboard of filtered metrics to our clients in real time? They have entrusted us with their business.
- *Process not tools*.

Demo – our app from earlier

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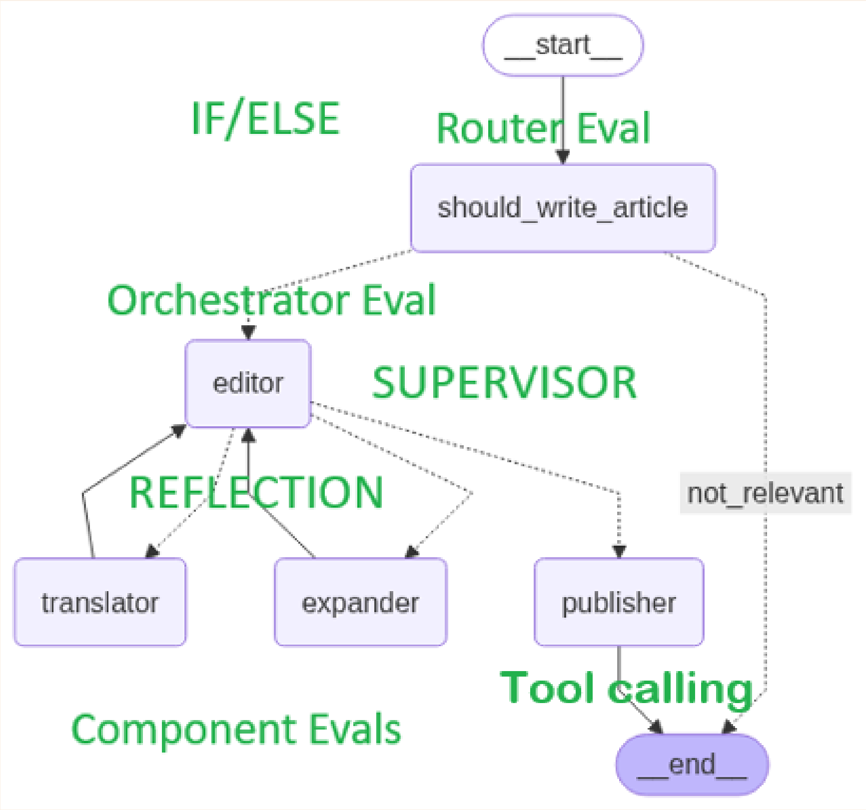

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Summary Key Points

- Evals are tests we carry out in the same way we would in tradional software development.
- We need to look at the data to determine what are EVAL criteria are
- We need to determine what success will look like
- We will need humans at certain points
- We need a Domain Expert
- LLM Judges are primarily for scaling
- We need to do EVALS on these LLM Judges
- Process more important than tools

Future

Craig West

https://craig-west.netlify.app/

https://evaluating-ai-agents.com/

Contact links on the sites

Always have 10 mins to answer questions...